

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			award (1) for similarity in results e.g. (both groups can draw the) same conclusions for samples A, C and D - volumes in the second experiment are approximately three quarters the values in the first experiment for samples A, C and D - results for samples A, C and D are reproducible award (1) for different conclusions drawn for sample B e.g. - first group found sample B to be permanently hard water but the second group found that it contains some temporary hardness and some permanent hardness			2	2		2
	(c)			calcium / magnesium ions <u>swap places with</u> / replaced by / exchanged for sodium ions (2) there must be one reference to <u>ions</u> if not complete answer award (1) for either of following calcium / magnesium ions removed / replaced sodium ions added if no mention of ions anywhere award (1) for calcium / magnesium swaps places with / replaced by / exchanged for sodium do not accept 'displace'	2			2		
	(d)	(i)		3.2×10^{-3} mol (2) award (1) for 0.0032		2		2	2	